RANJITH PALANI M

Senior Data & AI Solutions Engineer | Cloud Data Architect

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EXECUTIVE SUMMARY

High-impact **Senior Data & AI Solutions Engineer** with 4 years of specialized engineering excellence architecting enterprise Cloud Data Lakes, GraphRAG & Generative AI solutions, real-time Computer Vision testing harnesses, and executive C-Suite BI platforms across the automotive domain. Functionally leading an engineering sub-team of **3 members** and serving as **Cross-Project Technical SME / Solutions Architect across 2 additional high-visibility initiatives**—diagnosing systemic bottlenecks, restructuring pipeline architectures, and driving **GCP FinOps optimizations** (Cloud Run concurrency tuning, BigQuery slot/partitioning optimization) to **slash monthly cloud billing by 35%+** while upscaling system throughput. Handpicked by Division Executive Leadership for **dual international on-site R&D missions to Prague (Czech Republic)** to resolve high-stakes gemba data discrepancies, coach European test leads, and align executive release readiness reporting across Tier-1 automotive programs (Volkswagen, BMW, Honda). Rated **"Exceeds Requirements"** in the latest annual performance appraisal and two-time recipient of departmental excellence awards.

CORE TECHNICAL COMPETENCIES

Artificial Intelligence & GenAI: GraphRAG, Knowledge Graphs, Google Cloud Vertex AI, Large Language Models (LLMs), Multimodal Vision, Qdrant Vector DB, Text Embeddings, Prompt Engineering & Strict Hallucination Suppression, Scikit-learn, CNNs.

Computer Vision & Hardware Telemetry: OpenCV (HSV color segmentation, 60 fps temporal tracking), pyzbar, Vector CANoe Integration (Python TCP/IP socket server & CAPL automation), High-density QR/IMEI decoding, Bench camera telemetry.

Cloud Data Engineering & FinOps: Google Cloud Platform (GCP), Advanced SQL / BigQuery SQL (Analytical CTEs, Window Functions, Partitioning, Clustering, FinOps Cost Optimization), Cloud Run (FastAPI Microservices, Autoscaling & Concurrency Optimization), GCP Security & IAM (OAuth 2.0 Client IDs, Service Accounts, API Keys), Ingestion Pipelines (ETL/ELT), SQLite (Local Mass-Data Staging & High-Throughput Caching), REST & OSLC APIs.

DevOps, Tooling & Automation: Jenkins CI/CD (Multi-Node Distributed Farm, Groovy Pipelines), Selenium, Python (FastAPI, PyInstaller, Loguru, SQLite3), SQL Automation & Database Migration, Git/GitHub, Windows automation.

Executive BI & Decision Analytics: Looker Studio (C-Suite Governance, 5-Tier Attribute Dashboards), Power BI (Basics), Advanced SQL (Window Functions, Conditional Compliance Modeling), Google Workspace APIs.

Leadership & Domain Standards: Functional Team Leadership (3 engineers), Cross-Project Technical SME, Cross-Site International Delivery (Prague R&D Gemba), Automotive V-Cycle, ASIL-B Safety Standards, ISO 26262 Traceability.

PROFESSIONAL EXPERIENCE

Automotive Systems & Software R&D

Valeo India Private Limited

Chennai, India & Prague R&D, Czech Republic

System Engineer (Functional Team Lead & Solutions Architect — Data, AI & Cloud)

Jan 2023 – Present

1. Technical Leadership, Cross-Project SME & Cloud FinOps Optimization

- **Sub-Team Leadership & Engineering Mentorship:** Functionally lead and mentor a sub-team of 3 engineers, overseeing technical designs, conducting pull request code reviews, sprint task distributions, and establishing standardized Python and data engineering practices.
- **Cross-Project Technical SME / Solutions Architect:** Appointed as transversal Technical Consultant for 2 separate high-visibility organizational projects outside core deliverables; diagnosed root-cause architecture failures, restructured data pipelines, and resolved production roadblocks to accelerate release velocity.
- **GCP FinOps & Cloud Billing Reduction:** Spearheaded cloud cost optimization across GCP services: restructured BigQuery queries with day-partitioning and clustering to eliminate full-table scans; tuned Cloud Run CPU allocation and autoscaling concurrency; secured multi-project cloud services using IAM Service Accounts, OAuth 2.0 Client IDs, and API Keys; eliminated redundant ETL polling—slashing monthly cloud infrastructure billing by 35%+ while boosting query throughput.

2. Enterprise Cloud Data Warehouse & Automotive V-Cycle Traceability

- **Master Traceability Data Warehouse (500,000+ Entities across 400+ Projects):** Architected an enterprise BigQuery master traceability data warehouse unifying 500,000+ live engineering entities across the entire automotive V-Cycle (Customer Needs, System Requirements, Architecture Specifications, and Test Executions) for 400+ projects and 8 autonomous vehicle subsystems (Detection, Perception Vision, Radar, Ultrasonic Sensors, Surround View, Long-Range Cameras, Communication, Gateway/Decision).
- **Official C-Suite Executive BI Platform:** Engineered the primary executive Looker Studio governance platform adopted by Division Executive Leadership and Program Directors as the mandatory single source of truth for weekly release readiness and milestone tracking across European programs.
- **Dynamic Defect Compliance & Quality Analytics:** Engineered advanced BigQuery SQL analytical CTEs with conditional validation logic that dynamically evaluates test execution states before defect checks—eliminating legacy 'false-negative' quality penalties on unexecuted or cleanly passed tests and delivering accurate, unskewed defect escape rates and pass-rate KPIs for executive decision-making.
- **End-to-End ALM Traceability ETL Pipeline (DOORS to BigQuery to Looker Studio):** Architected an end-to-end ALM traceability pipeline extracting requirements and verification metrics from IBM DOORS, CodeBeamer, and Jira into BigQuery and Looker Studio—maintaining horizontal and vertical traceability across customer goals and components; modernized legacy spreadsheet syncs into a resilient Python-to-BigQuery automated engine with local SQLite staging for high-volume offline ingestion, slashing pipeline runtimes by 70%.

3. Generative AI, GraphRAG & Computer Vision Innovation

- **BigQuery GraphRAG Knowledge Base:** Pioneered a GraphRAG architecture directly inside Google Cloud BigQuery, structuring 8,500+ DOORS specification modules into semantic nodes and directed dependency edges to achieve deterministic, zero-hallucination requirement queries and automated traceability via Vertex AI LLMs.
- **Multimodal AI Vision & Hardware Identification:** Developed a standalone zero-dependency Windows vision application using PyInstaller and Vertex AI Multimodal Vision with strict anti-substitution prompt engineering, eliminating OCR character swaps (e.g., 0/O, 8/B) on physical ECU hardware labels during ISO 26262 bench testing setup and linking telemetry directly with Vector CANoe test benches.
- **Automotive Bench Computer Vision (OpenCV & CANoe Integration):** Engineered an automated real-time test harness validating instrument cluster LEDs (Honda Dome / Cluster); built OpenCV HSV color segmentation and 60 fps temporal tracking to verify high-frequency blinking cycles, linking verdicts to Vector CANoe CAPL test cases via a custom Python TCP/IP socket server.
- **AI Test Case Generation Platform & Cloud Microservices:** Architected an AI-powered automated test case generation platform using FastAPI microservices on Google Cloud Run, Qdrant vector database, and BigQuery; ingested raw stakeholder requirements from the traceability ETL pipelines and leveraged LLMs to auto-generate system test cases and CAN signal mappings—slashing manual test authoring time by 40%+ and accelerating engineering release cycles.

4. Launch Readiness Review (LRR) Full-Stack Governance Platform

- **Full-Stack Governance WebApp & Concurrency Control:** Developed a full-stack governance web application using Google Apps Script (HTML5/JavaScript frontend, modular GAS backend) integrated with BigQuery; engineered optimistic concurrency control (version tokens), an automated database snapshot freeze engine, and automated review milestone Jenkins dispatchers, eliminating write collisions across 50+ Project Leads and Technical Program Managers.
- **High-Availability Automation & Timesheet Compliance Bot:** Engineered a standalone Selenium desktop automation bot extracting resource hours and dispatching multi-tiered management reminders, saving dozens of operational follow-up hours monthly; maintained high-availability multi-node Jenkins CI/CD pipelines executing automated regression test suites with zero downtime.

5. International On-Site Missions & Global Leadership (Prague R&D Center, Czech Republic)

- **Dual International Missions (Feb & Oct 2026):** Handpicked by Division Executive Leadership for two on-site missions to Prague R&D Center as the sole technical specialist representing Chennai GTC.
- **Executive Reporting & Cross-Site Coaching:** Delivered executive readiness dashboards for Prague engineering directors, and conducted technical workshops and coaching for European test leads across multiple vehicle programs on automotive V-Cycle traceability, executive KPI reporting, and deploying the newly developed AI-powered test case generation platform.
- **High-Stakes ASIL-B Safety Validation & Anomaly Traceability:** Collaborated on-site with European test leads in the gemba to resolve critical ASIL-B safety test coverage shortages (e.g. ADAS Travel Assist autonomous driving), eliminating 100% of unmapped dashboard categories and connecting defect escapes directly to test execution records.

EDUCATION & PROFESSIONAL CERTIFICATIONS

Bharathiar University | Master of Science in Data Analytics — Coimbatore, Tamil Nadu | **2021 – 2023**

Madras Christian College | Bachelor of Science in Mathematics — Chennai, Tamil Nadu | **2018 – 2021**

- **Data Analysis with Python:** IBM Professional Certification
- **Database Management Systems:** NPTEL Certified

HONORS & RECOGNITION AWARDS

- **Performance Appraisal Excellence:** Awarded "Exceeds Requirements" in the latest annual performance appraisal; formally commended by management as an indispensable technical asset to the division.
- **Departmental Recognition Awards (2026):** Two-time recipient of the "Rising Star" and "Shining Star" awards in 2026 for technical excellence, cross-site leadership, and high-impact delivery.
- **Smart India Hackathon Finalist (SIH 2022):** Developed an AI-driven personalized dashboard for identifying and supporting special-needs slow learners using Python, Streamlit, and Power BI.
- **Accenture Data Analytics Virtual Experience:** Analyzed Social Buzz social media datasets to identify top content categories for advertising targeting using Excel.